

# Architecting a Profession

## The Roadmap Journey

### Abstract

Enterprise Architecture is becoming increasingly important for organizations dealing with accelerating levels of change amidst a more complex (and often uncertain) number of variables. While thousands of people worldwide are “practicing” as Enterprise Architects, there is no commonly accepted baseline of knowledge nor are there standards and guidelines to ensure consistency of service delivery within the context of a profession. Formalization of a recognized profession of Enterprise Architecture requires the development of the elements of a profession agreed upon by the body of practitioners. In the past, formal professions have gradually evolved over time. In recent years, considerable study has been given to the question of what constitutes a profession.

Most recognized professions evolve to their current state over many years (decades and longer in some cases). The challenge is to present a framework to accelerate the cycle time for building and achieving a steady state for a new profession (Enterprise Architecture in this case) to address the growing demand for reliable and consistent delivery of services. The IEEE Computer Society developed a model of the elements of technical professions that has been used as the basis for a roadmap to architect the profession of Enterprise Architecture.

### Why Establish Enterprise Architecture As a Profession?

The most critical issues facing enterprise architecture are i) an almost total lack of formal, recognized governance, ii) significant inconsistency in practices, and iii) inability to provide assurance that functionally equivalent results can be delivered by all professional enterprise architect practitioners. Today, the practice of enterprise architecture is inconsistent, and lacks a commonly agreed upon definition and understanding. This fractionalization has hurt the legitimacy of qualified practitioners since anyone can call himself or herself an enterprise architect, as there are no standards from which to measure and assess basic levels of competency. Rather, views of Enterprise Architecture are typically constrained by political, cultural, and social elements within each enterprise. Enterprise Architecture does not yet have the professional standing found in well-established professions like medicine, law, engineering, and accounting.

The goal of pursuing a profession now is to address these concerns so that Enterprise Architects can consistently deliver value to enterprises that are in the midst of accelerating levels of change in increasingly technical and complex environments. More and more organizations are turning to Enterprise Architecture to help plan, design, and augment the execution of change to ensure they can survive and thrive amidst increasing uncertainty.

## Why now?

As in the early days of older professions, many grass-roots organizations have sprung up to support practitioners as they struggle to reach agreement about Enterprise Architecture practices, artifacts, and benefits to the enterprise. Until now, there has been no unifying structure to bring them together.

Now, however, the Federation of Enterprise Architecture Professional Organizations (FEAPO), a worldwide association of professional organizations, provides a forum to standardize, professionalize, and otherwise advance the profession of Enterprise Architecture. FEAPO provides a focused hub for the Enterprise Architecture expertise and resources of its member organizations, which include, in addition to the IEEE Computer Society and the Center for the Advancement of the Enterprise Architecture Profession, such organizations as

- The Association of Business Process Management Professionals,
- The Business Architecture Guild,
- The Business Architecture Society,
- The Canadian Information Processing Society,
- Data Management International (DAMA), and
- The International Council on Systems Engineering (INCOSE) .

Prior to the establishment of FEAPO, the fragmented efforts failed to achieve a commonly agreed definition of a well-established and consistent set of Enterprise Architect capabilities or a clear differentiator between Enterprise Architecture and the many flavors of “Architecture” in the enterprise and its information technology component. Now, FEAPO’s member organizations have come together to guide the nascent profession to a self-governed status of maturity by 2017, to progress from organized, to qualified, to self-governed, with increasing maturity in three distinct areas:

1. Standards of practice
2. Professional learning
3. Industry self-governance and concomitant professional autonomy

## The First Step: Building Blocks of a Profession

What is the essential distinction between a “line of work” and a “recognized profession”? This is a question that the IEEE Computer Society has previously addressed in the formation and recognition of Software Engineering as a profession. As a founding member of the Federation of Enterprise Architecture Professional Organizations (FEAPO), the Computer Society is sharing its experience and its model of a profession with FEAPO. Significantly, the Center for the Advancement of the Enterprise Architecture Profession (CAEAP), another founding member of FEAPO, has also done extensive work in defining Enterprise Architecture as a profession. Both organizations have identified similar hallmarks of a profession including the need for a structured educational program. Fortuitously, during the same timeframe, Penn State University began addressing the need for undergraduate, graduate, and continuing education curricula to educate students and practitioners in the field of Enterprise Architecture.

Some practitioners do not see a need to professionalize Enterprise Architecture as rigorously as medicine, law, engineering, accounting or even software engineering. Others are in favor of professionalization to the fullest extent. This debate is centered on whether to position Enterprise Architecture as a legally recognized profession or as a generally accepted profession.

If Enterprise Architecture is to be recognized as a generally accepted profession, it sets its own standards and governs itself, at a minimum, by the four E's:

1. Education
2. Examination
3. Ethics
4. Experience

If Enterprise Architecture is to be recognized legally as profession, it must address additional concerns, such as government certification or licensing by state authorities.

Interactions among the above topics are more complex than they might seem at first glance. For example, the requirements for professional licensing can have a significant impact on the content of initial education. Certification guidelines can influence the content of professional development materials. A professional society is required to manage the certification process and develop an appropriate code of ethics.

Whether legally recognized or generally accepted, the following common traits are shared by both types of professional status:

1. A body of knowledge founded on well-developed and widely accepted theoretical and practical bases.
2. A system for certifying that individuals possess such knowledge before they enter practice.
3. A progressive system of certifying practitioners as they increase both their knowledge and experience in effective practice.
4. A code of ethics, with a commitment to use specialized knowledge for the public good.
5. A professional society, with provisions for monitoring individual compliance to ethical standards and professional practices.

The model that FEAPO has adopted for accelerating the recognition of Enterprise Architecture as a recognized profession includes the elements needed for both: the full model reflects the elements needed for a legally recognized profession; omission of the licensing element is appropriate for a generally accepted profession.

## **Step 2: Assembling the Building Blocks into a Model of the Profession**

FEAPO has ratified the model below as an Organization Technique to guide the sequence and investment of effort required to “architect” the practice of Enterprise Architecture as a recognized

profession. The individual elements are shown in Figure 2 and described below in clockwise order, as shown by circled numbers 1 - 7.

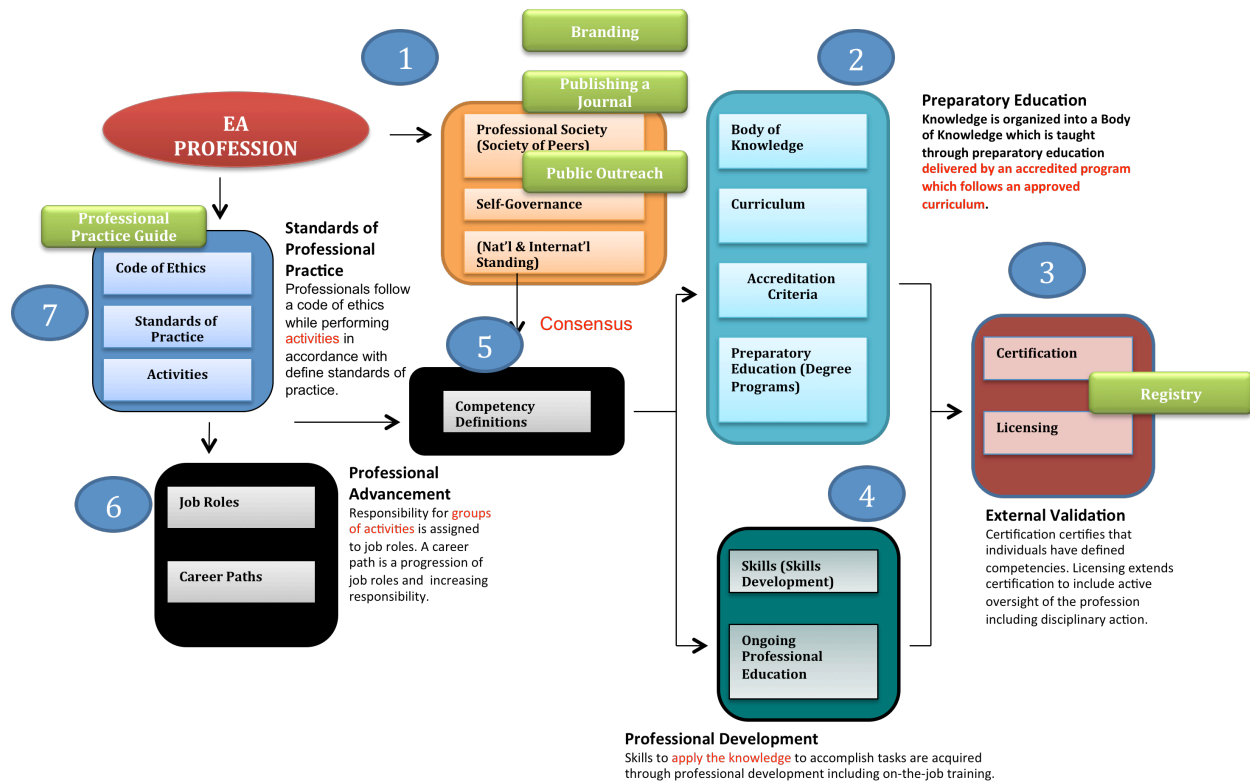


Figure 1: Blended Model of the Enterprise Architecture Profession (IEEE CS + CAEAP Elements)

### Professional Society

An international professional society supports the world-wide advancement of the profession through such means as branding, publishing journals, and various forms of public outreach. The society works for sufficiency and sustainability of employment in the profession, and promotes active research efforts to advance the state of the profession's knowledge. It provides a mechanism for self-governance of the profession.

### Body of Knowledge

While the body of knowledge (BOK) for enterprise Architecture is vast and its many parts reside in many books and websites, the professional society should assure the availability of a guide to that body of knowledge. The Guide clearly defines the core content and boundaries of the discipline (what's in and what's out) and can point to specialized bodies of knowledge beyond "core," and can identify supporting knowledge areas such as technical management and enterprise operations. The Guide should obtain consensual validation of the knowledge, and provide a reasoned foundation for the knowledge.

### Curriculum Model(s)

A curriculum model provides educational institutions with a common basis for the establishment and improvement of educational programs (curricula), and thus should be defined by recognized, authoritative bodies, use the profession's Guide to its Body of Knowledge as a source of information, and be aligned with the Competency Model. Since the Enterprise Architecture BOK is extensive, different curriculum models are needed for undergraduate and graduate programs.

### **Accreditation Criteria**

An accreditation system assures the quality and suitability of preparatory education. Accreditation is a public statement of an institution's capacity to deliver effective learning programs based on consensually agreed criteria. For professions that require practitioners to be certified, accreditation criteria should be aligned with certification requirements.

### **Preparatory Education**

Preparatory education programs must be readily available, and comply with approved curriculum models and accreditation criteria. An Enterprise Architecture program should provide the education and training necessary to be employed in an entry level position in the profession. For ABET<sup>1</sup> certification, "The program must have and enforce policies for accepting both new and transfer students, awarding appropriate academic credit for courses taken at other institutions, and awarding appropriate academic credit for work in lieu of courses taken at the institution. The program must have and enforce procedures to ensure and document that students who graduate meet all graduation requirements."

### **Certification**

Certification represents validation, by a community of peers, that an individual possesses the knowledge and competence of a professional. Meaningful certification requires a standard and recognized certification regime that is consistent with the profession's body of knowledge and its competency model. Mechanisms for competency certification should not require compliance with education requirements (i.e., they should enable recognition of competency acquired by other means). Thus, it needs to include demonstration of competency through practice (e.g., apprenticeship / experience). There should be a public registry of certified professionals, along with a defined means of re-certification/CPD & de-certification of individuals.

### **Licensing**

While certification is often managed by professional societies, licensing is implemented by other authoritative bodies, such as an agency of one's state of residence or the state of work. Like certification, licensing must be supported by a recognized body of knowledge, certification standards, competency standards, code of ethics and a clear boundary of a licensed discipline, and result in a public registry of licensed professionals

### **Skills**

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<sup>1</sup> Accreditation Board for Engineering and Technology

Skill is the ability to do something well; to take what you know and apply it so as to cause the desired effect to occur<sup>2</sup>. The ability comes from aptitude and practice of the application of one's knowledge.

### **Professional Development**

Professional development opportunities may come from multiple sources of education and training that enable the advancement of skills and knowledge to succeed in the profession of Enterprise Architecture. Increasing experience in the practice of Enterprise Architecture is also necessary to keep current and to advance one's expertise in the profession.

### **Competency Model**

A competency model provides a unified view of the knowledge, skills and levels of competence required of professionals by providing definitions of knowledge and skills at increasing levels of maturity in professional development. It must be accompanied by an ongoing maintenance program to update the model on a regular schedule.

### **Job Roles**

Job roles<sup>3</sup> are specified in terms of responsibilities for the role's required activities and in terms of the competencies required to perform the activities. Role descriptions should clearly delineate the level of responsibility(ies) that the role is accountable for.

### **Career Paths**

"Career path" refers to one's progression in a profession where successful experience in previous job roles is necessary to subsequent job roles. Successive job roles are expected to require higher levels of competency.

### **International Standards of Practice & Technical Standards**

Defined and recognized professions are characterized by known standards of practice for performing activities in one's job role. A standard of practice represents an acceptable level of performance or an expectation for professional intervention, formulated by professional organizations based upon current knowledge and expertise. Such information may be captured in a Professional Practice Guide, such as that provided by CAEAP. Technical standards specify techniques, methods, procedures and performance norms, consensually agreed by the profession. Such standards often specify the form and content of work products and appropriate metrics.

### **Defined Activities**

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<sup>2</sup> A certain amount of knowledge is a prerequisite of skill; you can't be skillful without first being knowledgeable; however, you can easily be knowledgeable without being very skillful.

<sup>3</sup> A job (position) description is different than a job role. A position description usually includes an explanation of where the position fits in the organization, such as to whom the position reports. A position description may contain

Specification of an Enterprise Architecture Job Role is facilitated by the existence of a known set of activities required in the performance of an Enterprise Architect's work. Within the profession, activity names are commonly understood and can also be used in position descriptions.

**Code of Ethics**

Most professions have written codes of ethics that are readily available to the public. A Code of Ethics specifies appropriate professional conduct to engage in the profession, including guidance for using specialized knowledge for the common good. (For example, customers may be the initial consumers of the Enterprise Architect's services, but the professional Enterprise Architect must always ensure the needs of the public are not sacrificed at the expense of the customer.)

**Step 3: How do we “build” the Profession?**

The elements discussed above provided us with the elements of a vibrant and generally recognized profession. The next step involved defining a roadmap to build out the elements using a phased approach (see Figure 3 below), aligned with key milestones. A number of concurrent activities are currently in progress, as FEAPO member organizations work together to accelerate the development of the various elements that are needed to unify and support Enterprise Architecture as a profession.

Each of the individual members have previously developed artifacts that contribute to the elements, and are now using the roadmap to prioritize their efforts to bring the disparate artifacts into a coherent whole.

## Overview of the Proposed Roadmap

### Maturity Milestones

Inception

Foundation

Education

Acknowledgement

Registration

Profession

Maintenance

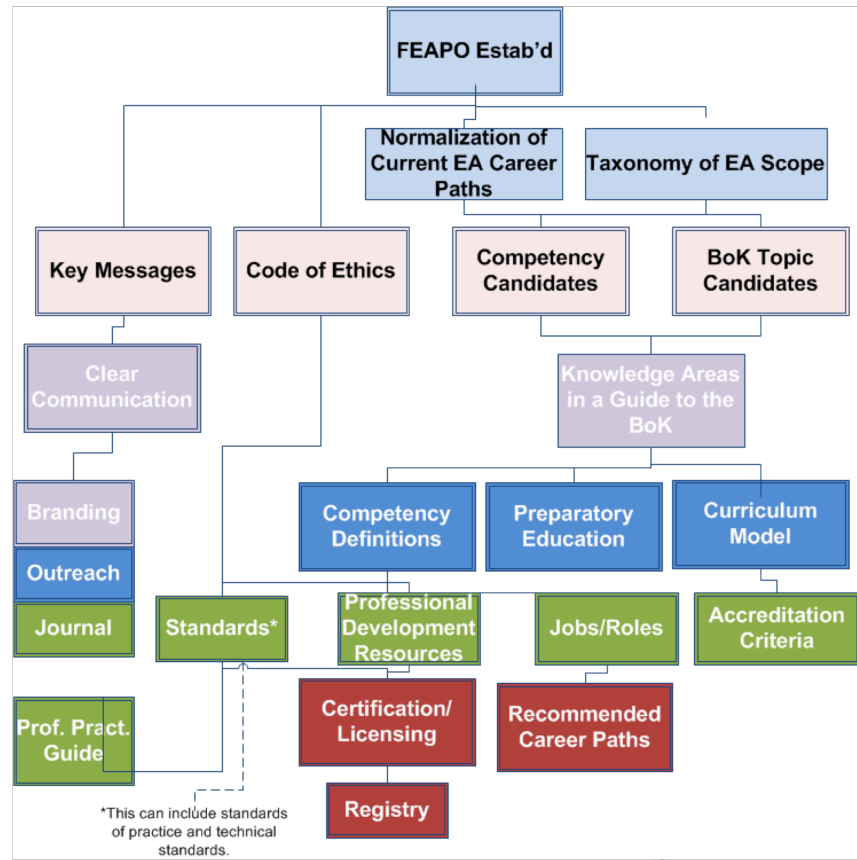


Figure 2: Overview of the Enterprise Architecture Roadmap

The first step in the roadmap has already been accomplished: FEAPO has been established, and includes the key players from the set of organizations serving Enterprise Architecture practitioners. These member organizations have individually established valuable artifacts and stores of knowledge supporting the practice of Enterprise Architecture. FEAPO's challenge is to build upon those in order to achieve a single profession. Achieving this goal necessarily involves collaboration in harmonizing the elements that already exist.

Figure 4 shows the timeline for this work. It shows, for example, that before we can achieve agreement on what belongs in the Guide to the Enterprise Architecture Body of Knowledge (EABOK), we need to agree on the scope ("taxonomy") of Enterprise Architecture — what's in and what's out. In order to agree on the knowledge areas to be described in the Guide, we should have agreement on what competencies a skillful Enterprise Architecture practitioner demonstrates. These efforts can be developed in tandem with (and will feed into) a list of topics for the EABOK. The competencies and topics can then be sorted into Knowledge Areas (KAs) to be addressed in the EABOK. Similarly, by examining existing Enterprise Architecture career paths from corporations and government bodies, we can validate the posited competencies and knowledge areas.

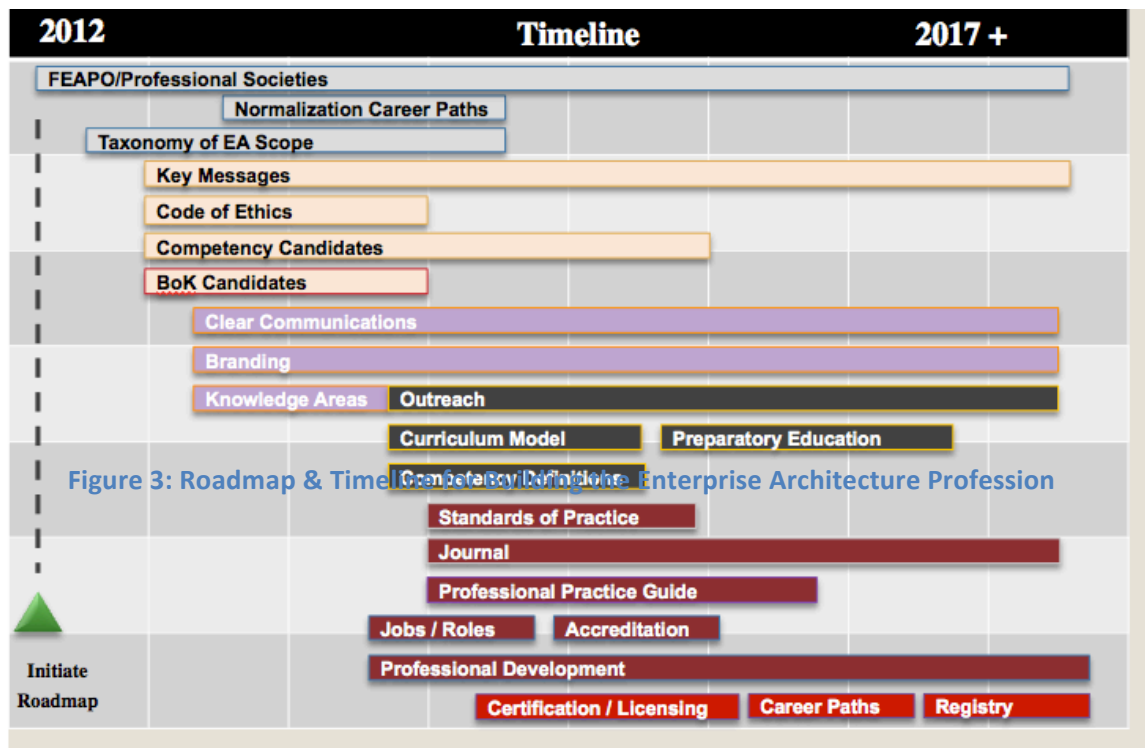


Figure 3: Roadmap & Timeline for the Enterprise Architecture Profession

Agreement on the KAs facilitates agreement on how to use the FEAPO Member Organizations' (FMO) existing knowledge bases and artifacts. Some may be folded into or otherwise support the consensually agreed KAs in the EABOK. In other cases, the EABOK may cite an FMO's artifact(s) as a primary reference. In any case, the agreed KAs provide a basis for commonly accepted definitions of requisite competencies and thus provide a basis for developing a Model Curriculum. The Model Curriculum provides a basis for educational institutions to develop their own specific preparatory education courses.

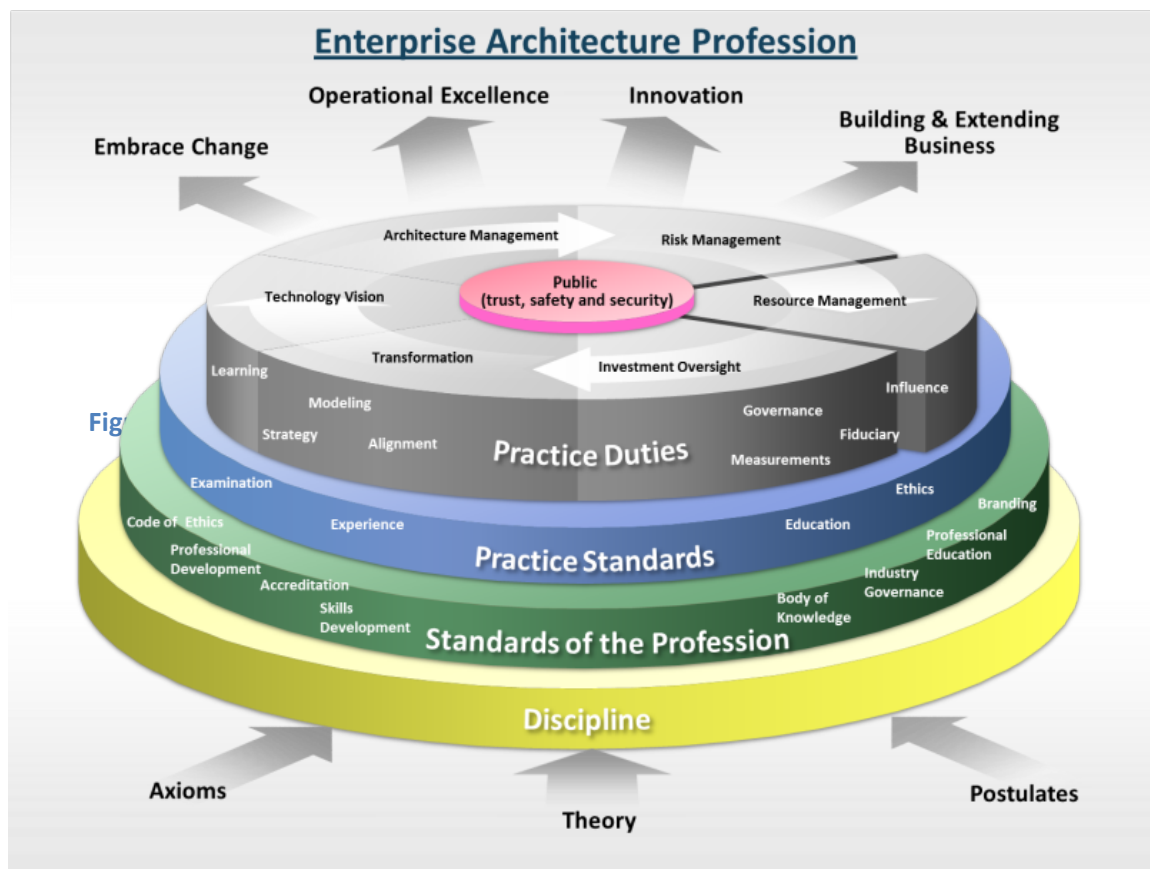
Development of a competency model assembles the requisite competencies in increasing levels of competency. This lays the foundation for grouping competencies into prototypical job descriptions. The job descriptions can be used for laying out archetypes for possible career paths.

## What Will the Unified Profession of Enterprise Architecture Look Like?

The profession will capture a skilled Enterprise Architect's portfolio of knowledge, skills, and experience and validate it via a rigorous qualification process to ensure truth and transparency, and to promote trust. It will enable that portfolio to be transferred to future generations of Enterprise Architects who will improve upon it. In addition, the profession will support practitioner career goals by providing guideposts for practitioners who want to achieve professional status and by recognizing individual achievements along the road to becoming fully capable professionals

The figure below depicts an Enterprise Architecture profession that delivers value across four dimensions:

1. Embracing Change by implementing the right structures and processes to reduce friction associated with change (across the business planning, operational, and technical areas).



2. Enabling Operational Effectiveness, going beyond simple efficiency, by ensuring that business assets (including human ones) are utilized properly and for the right purposes.
3. Enabling sustainable innovation by assuring that the right structures are in place across the organization to manage the multiple portfolios that need to align and work synergistically.

4. Building & Extending the Business (organization), by focusing on alignment of strategy, risk, culture, and capabilities to extend an organization's ability to deliver sustainable value. The profession is built around a core focus on serving the public.

This foundation for the profession supports creation of the right skills, capabilities, and knowledge to ensure consistent and ethical delivery of services, as well as the necessary oversight to assure the needs of the public are addressed and standards for the profession are upheld via measurement, certification, and governance.

## The Road Ahead | The Journey Continues

The journey towards building the profession of Enterprise Architecture begins like all others with the first step. FEAPO has taken that first step, and FEAPO Member Organizations are now working together to:

1. Instantiate the requisite elements of a profession described above
2. Promote the art and science of Enterprise Architecture by uniting likeminded individuals in the major organizations of the Enterprise Architecture ecosystem
3. Identify and establish a body of ethics, principles, and maturity measures that support the Enterprise Architecture profession and its practice.
4. Develop and certify a body of knowledge that properly represents the Enterprise Architecture Profession and its practice.
5. Improve the practice of Enterprise Architecture by providing consensually based artifacts, education and certification
6. Establish collaboration for skills development and professional development between professional societies
7. Establish collaboration avenues for professional education and model curricula developers to achieve accreditation.

We invite enterprise architects and partners with an interest in the journey to join the discussion and participate in the building of this new profession.



*"Perhaps no human artifact is more important to the survival of the species than the organization of enterprises. Nearly everything humans do together is done in the context of some kind of enterprise – families, businesses, governments, social groups, educational institutions, military organizations, charities, and so on. Relatively small, static, or simple enterprises in general may not require an explicitly documented enterprise architecture. But most enterprises of consequence are large, dynamic, complicated, and important. Random, organic, component-focused growth and management are no longer viable options for enterprise management. The current global economic situation is strong*

*evidence of this. Enterprises must be architected and managed by that architecture if they are to remain viable in the Information Age. The Enterprise Architecture profession can make that happen.”<sup>4</sup>*

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<sup>4</sup> “To Be or Not to Be: Recognize Enterprise Architecture as True Profession?,” by Mark Lane, in *The SIM Guide to Enterprise Architecture*, edited by Leon Kappelman, CRC Press, 2009, p. 60.